

INVESTIGATION REPORT

Issue 1010: Debug single-step inside a hardware loop

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1010
Disposition	Dead code, proven
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1010
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	COREV_PULP=1 FPU=0
Disposition	Dead code, proven
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	5 cited, 4 present, 1 not retained

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The branch at cv32e40p_controller.sv:850 (debug_single_step_i & ~debug_mode_q inside DECODE_HWLOOP) and its dependent lines 852-887 are unreachable. An unbounded proof (abc pdr, top=cv32e40p_core, COREV_PULP=1) shows that no reachable state has ctrl_fsm_cs==DECODE_HWLOOP together with debug_single_step_i=1 and debug_mode_q=0; a second proof of the exact line-850 guard (including instr_valid_i, no pending debug request/trigger, no IRQ) also passes. The enclosing DECODE_HWLOOP branch itself is reachable (bounded cover hits it at step 8), so the proofs are not vacuous. Mechanism: dcsr.step can only be written in debug mode (decoder makes dcsr access illegal otherwise); leaving debug mode happens only via dret through XRET_JUMP which lands in DECODE; in DECODE with step=1 the line-664 logic halts IF and redirects to DBG_FLUSH before DECODE_HWLOOP can be entered, and the proof confirms the id_ready_i=0 corner does not open a path. Environment assumptions: OBI handshake legality, scan_cg_en_i=0, fetch_enable_i=1.

3 Conditions

- 1. OBI handshake legality
- 2. scan_cg_en_i=0
- 3. fetch_enable_i=1

Extracted verbatim from the environment assumptions stated in the verdict record for this run.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_controller.sv:850-887 (DECODE_HWLOOP single-step branch): formally proven unreachable. dcsr.step is writable only in debug mode; debug mode is exited only via dret (XRET_JUMP -> DECODE); in DECODE with step=1 the controller halts IF and routes to DBG_FLUSH/FLUSH_EX/DBG_WAIT_BRANCH, so DECODE_HWLOOP is never entered with step=1 outside debug mode. Proof: runs/20260909-1238-cv32e40p-1010/formal/p_assert (abc pdr, unbounded) + anti-vacuity cover c_state. Exclude branch and conditions at line 850 and lines 852-887 from coverage.

Checker used

checker_1010.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/p_assert/sby.log	present	abc pdr PASS (48.8 s): exact guard of line 850 never true in any reachable state
E2	proof	formal/p_strong/sby.log	present	abc pdr PASS (39.2 s): DECODE_HWLOOP never coexists with debug_single_step_i=1 & debug_mode_q=0
E3	proof	formal/c_state/sby.log	present	smtbmc cover PASS (33.6 s): DECODE_HWLOOP state and the exact enclosing branch reached at step 8 (covers_reached=2) -> asserts not vacuous
E4	counterexample	formal/c_state/job/engine_0/trace1.vcd	not kept	cover witness trace: entry into DECODE_HWLOOP via hwloop body; rendered in cover_decode_hwloop.png
E5	sim-log	sim/hwloop_dt/sim.log	present	Directed test hwloop_dt (real cv.setupi hardware loop, 10 iterations) on the core-v-verif Verilator testbench: EXIT SUCCESS, acc=30, controller observed in DECODE_HWLOOP in simulation with debug_single_step_i=0 (sim_hwloop_witness.png)

Paths are relative to runs/20260909-1238-cv32e40p-1010. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- sim-log. Simulation. Exercises the stimulus that was run. It does not establish an unbounded proof.

Artefact(s) E4 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- The proof establishes unreachability under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

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Provenance and measurement

Run directory	runs/20260909-1238-cv32e40p-1010
Title source	Title and configuration read back from reports/issues/cv32e40p_issue_1010.pdf, the previously published report for this issue.
Inputs shown	title, body(links stripped), labels
Human record	Avinash 2026-09: yosys/abc PDR unbounded proof (~100 s), controller never in DECODE_HWLOOP while single-stepping outside debug mode
Agent confidence	0.9. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.
Iterations	1

Measured tool and step time

Step	Measured
fetch issue (blind)	12.0 s
locate line, signals, drivers	60.0 s
classify + write checker	63.0 s
proof p_assert (pdr)	48.8 s
proof p_strong (pdr)	39.2 s
cover c_state (smtbmc)	33.6 s
write verdict + report + PDF	2.1 min (128 s)
directed test (compile + Verilator, waves)	20.1 s
Total	6.4 min (384 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

Note. phase times derived from log timestamps 12:38:35 -> 12:44:59 and per-job engine seconds

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